

15 阶三次单位根最大行列式：最后三层与全局上界

$Q = 96, 99, 105$ 的完整排除·38 / 38 ·精确计算机辅助证明

研究续篇与可复演证书

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1. 结论、范围与证明组织

令 $\omega^2 + \omega + 1 = 0$, $\mu_3 = \{1, \omega, \omega^2\}$, 并设

$$G = HH^*, \quad K = H^*H, \quad Q = \sum_{i < j} |G_{ij}|^2, \quad H \in \mu_3^{15 \times 15}.$$

沿用原独立审计 [A17]、35 层续篇 [P35] 中的结构证明，以及已发表的 $B_3(15, 10) = 12$ 码界 [TB20]，本续篇补齐 $Q = 96, 99, 105$ 。合并这些结果得到

$$\max_{H \in \mu_3^{15 \times 15}} |\det H|^2 = 277868041444786176 = 2^{22} \cdot 3^{20} \cdot 19.$$

等价地，最大行列式模为 $120932352\sqrt{19}$ 。原纪录矩阵达到这个值，故上界与下界相合。本结论确定最大值；没有声称已经分类所有等号矩阵，也没有声称最大化矩阵在通常等价关系下唯一。

这是带有精确有限证书的**计算机辅助数学证明**。本报告说明结构覆盖和各排除判据的数学依据；程序完成有限候选的生成、整数与有理运算、两种独立的全量列枚举，以及证书核验。它尚未被翻译为 Lean、Coq 等证明助手的形式化推导。本次没有重新执行 [TB20] 的码分类。

[P35] 的原文档、历史 35/38 状态表和原执行入口保留不变。最终状态由新的 `run_final.py` 和 `results/final_ledger.json` 汇总，避免把旧报告中的研究缺口直接改名成定理。

2. 从 35 层到三个有限问题

[A17,P35] 已排除 $Q \not\equiv 0, 6 \pmod{9}$ 的情形，并证明所有 $Q \geq 171$ 的行列式平方小于纪录 B 。在剩余 38 个允许壳层中，35 个已完成。因此只需排除满足 $|\det H|^2 > B$ 的 $Q = 96, 99, 105$ 矩阵。

严格反例的行、列颜色型各自只可能为 (15)、(14, 1)、(13, 2) 或 (13, 1, 1)。对于 $Q = 96, 105$ ，只可能出现 (14, 1) 或 (13, 2)；对于 $Q = 99$ ，只可能出现 (15) 或 (13, 1, 1)。同色非零 Gram 项的平方模至少为 9；异色项平方模至少为 3。

把含 13 行的多数颜色主块记为 A ，另两行称为枢纽。令其内部能量、交叉能量和枢纽之间内积的平方模分别为 e, c, h 。先前的精确分量—Schur 界将三层的 size-13 余项限定为 $e = 0, 9, 18$ 。 $e = 0$ 会给出 13 个正交三次单位根行，由 [TB20] 排除； $e = 9$ 由 [P35, §9] 的单边缺陷引理排除。对于 $e = 18$ ，只剩

$$(e, c, h) = (18, 78, Q - 96).$$

两条内部边的平方模均为 9，故支撑只能为 P_3 加 10 个孤立点，或 $2K_2$ 加 9 个孤立点。此次没有删去内部孤立点，也没有使用已经失效的行列颜色“同方向轨道”论证。

本续篇先排除所有 size-13 余项。随后, $Q = 96, 105$ 的严格反例被迫在两侧都具有 $(14, 1)$ 型, $Q = 99$ 被迫在两侧都为全同色。这一顺序保证了跨颜色配对的覆盖; 不需要预先假定行 Gram 和列 Gram 处于同一颜色型。

3. 必须满足的列方程与精确穷举

列方程引理。 若可逆矩阵 H 满足 $HH^* = G$, 则

$$H^*G^{-1}H = I.$$

因此每一列 v 都必须属于

$$\mathcal{C}(G) = \{v \in \mu_3^{15} : v^*G^{-1}v = 1\}.$$

如果 $\text{span}_{\mathbb{C}} \mathcal{C}(G)$ 的维数小于 15, G 就不能由三次单位根矩阵分解。这个判据只利用一个 Gram 矩阵, 不需要知道另一侧的颜色、支撑或相位。

列方程在整体乘三次单位根下不变, 因此可以固定 $v_1 = 1$, 完整检查 $3^{14} = 4,782,969$ 个向量。设 $L > 0$ 清除逆矩阵全部分母, $J = LG^{-1}$ 。程序使用 $\mathbb{Z}[\omega]$ 中的整数系数, 检查

$$v^*Jv = L.$$

第一种算法按三进制次序改变坐标, 利用

$$q(v + de_k) = q(v) + 2 \operatorname{Re}(\bar{d}(Jv)_k) + J_{kk}|d|^2$$

更新二次型, 并同步更新 Jv 。第二种算法为每个向量重新计算全部 105 个坐标对的二次型贡献, 完全不使用增量更新。两种算法在本次全部 194 个 Gram 候选上, 逐项得到相同的完整列列表。

每个候选还以任意精度 Eisenstein 运算检查 $GJ = LI$ 、行列式、能量和颜色同余。Numba 内核使用有符号 64 位整数; 令 M 为 J 两个整数系数的最大绝对值, 执行前检验 $100 \cdot 15^2 M < 2^{63}$ 和 $L < 2^{63}$, 这是涵盖上述更新及直接求和中间量的保守界。

对被列空间排除的候选, 程序额外提供一个明确的非零 $w \in \mathbb{Z}[\omega]^{15}$, 并在完整允许列列表上检查 $w^*v = 0$ 。因此最终证据包含 193 个可直接核验的湮灭向量, 排除依据不限于一个浮点秩或未经展开的 $\text{rank} < 15$ 返回值。

4. size-13 的规范化与候选覆盖

记 $\delta = 1 - \omega$, 行颜色为 r_i 。同一个 Gram 内的正确同余是

$$G_{ij} \equiv (r_j - r_i)\delta \pmod{(3)}.$$

可整体共轭, 使多数颜色与第一枢纽的颜色差为 1。由于所有交叉项平方模都是最小值 3, 可以仅用三次单位根行缩放, 将多数行到第一枢纽的项全部规范化为 δ 。第二组交叉项写为 $\varepsilon \delta t_i$, 其中 $t_i \in \mu_3$, $Q = 99$ 时 $\varepsilon = -1$, 另外两层 $\varepsilon = 1$ 。再用第二枢纽的三次单位根缩放固定一个核心坐标的 $t_i = 1$ 。

此处保持了 H 的三次单位根字母表。内部两条边分别枚举 $3\mu_6$ 中的全部六个值, 不把六次单位根行缩放误当成保留三次字母表的等价操作。对于 r 个内部孤立点, 只需记录 (n_0, n_1, n_2) , 其和为 r ; 相同取值坐标间的排列属于允许的行置换。

第一枢纽到第二枢纽的项在 $Q = 96$ 时为零; $Q = 99$ 时枚举 $\delta\mu_3$; $Q = 105$ 时枚举全部平方模 9 的 Eisenstein 整数。核心边值、核心比值、孤立点计数和枢纽项均由有限循环生成。每个生

成的 Gram 先接受精确 Schur 行列式计算, 再接受 $D > B$ 、整数、 $3^{14} \mid D$ 及 Eisenstein 范数条件。

这些条件都是真实因子的必要条件。 $3^{14} \mid D$ 来自对 H 的 14 行减去第一行后提取 δ^{14} ; 一个非负整数为 Eisenstein 范数, 当且仅当每个 $p \equiv 2 \pmod{3}$ 的有理素因子的指数均为偶数。生成程序直接使用这些判据, 旧候选数值只用于一致性断言, 未作为枚举输入来预先删去未知数值。

壳层	P_3 规范代表	$2K_2$ 规范代表	合计	列空间排除
96	20	44	64	64
99	20	47	67	67
105	8	3	11	10
合计	48	94	142	141

“规范代表”允许仍有等价冗余; 表中数量并非不等价矩阵的分类数。所有 142 个候选都执行了两种全量列检查。唯一未被列空间维数排除的代表属于 $Q = 105$, 下一节给出其解析矛盾。

5. $Q = 105$: 模 4 范数矛盾与 $16 > 15$

最后一个 Gram 代表可写为

$$G_* = \begin{pmatrix} A & z & z \\ z^* & 15 & 3 \\ z^* & 3 & 15 \end{pmatrix}, \quad A = \text{diag}(C, C, 15I_9), \quad C = \begin{pmatrix} 15 & 3 \\ 3 & 15 \end{pmatrix}, \quad z = \delta \mathbf{1}_{13}.$$

其行列式为 $281238577200000000 > B$ 。它确实是一个必须进一步排除的抽象候选, 单纯把数值与纪录比较无法结束证明。

设一个可能列向量为

$$v = (a, b, c, d, p_1, \dots, p_9, u, v_0)^T \in \mu_3^{15}, \quad s = a+b+c+d, \quad P = \sum_{j=1}^9 p_j, \quad k = s/18 + P/15.$$

由 Schur 补精确求逆得

$$v^* G_*^{-1} v = x^* A^{-1} x + \frac{15}{392} |u + v_0 - 2\bar{\delta}k|^2 + \frac{1}{24} |u - v_0|^2,$$

其中 x 是前 13 个坐标。程序还以任意精度系数逐项核对这个二次型恒等式与完整逆矩阵一致。

假设同时有 $a \neq b$ 和 $c \neq d$ 。每个不同符号对在 C^{-1} 上的贡献为 $11/72$, 另外九个坐标贡献 $9/15$, 从而

$$x^* A^{-1} x = 2 \cdot \frac{11}{72} + \frac{9}{15} = \frac{163}{180}.$$

如果 $u \neq v_0$, 则 $|u - v_0|^2 = 3$, 上式立即给出

$$v^* G_*^{-1} v \geq \frac{163}{180} + \frac{3}{24} = \frac{371}{360} > 1,$$

与列方程矛盾。如果 $u = v_0$, 列方程则强制

$$\frac{15}{98}|u - \bar{\delta}k|^2 = \frac{17}{180}.$$

清除分母并利用 $\delta\bar{\delta} = 3$, 得到

$$N(30\delta u - 5s - 6P) = 1666.$$

左侧是 Eisenstein 整数的范数。对于 $N(a + b\omega) = a^2 - ab + b^2$, 模 4 的可能余数只有 0, 1, 3, 而 $1666 \equiv 2 \pmod{4}$ 。因此这一情形也不可能。两种枢纽情形都被排除, 故每个可用列必须满足

$$a = b \text{ 或 } c = d.$$

这条限制具有完整的代数证明, 独立于列枚举。作为实现交叉检查, 全部 5887 个规范化允许列中, 两对同时不同的列数确为零: 仅第一对不同为 2304 列, 仅第二对不同为 2304 列, 两对均相同为 1279 列。

现在回到整个矩阵。 $G_{*,12} = G_{*,34} = 3$ 。若一对长度 15 的三次单位根行在 a_0 个位置相同, 则其实内积为 $a_0 - (15 - a_0)/2$ 。实内积为 3 强制 $a_0 = 7$, 即每对行恰有 8 个不同位置。上面的列限制要求这两对行的不同位置互不相交; 15 个位置却需要容纳 $8 + 8 = 16$ 个位置, 矛盾。

所以 G_* 无法分解, $Q = 105$ 的全部 size-13 分支排除。[P35] 已排除该层双 (14, 1) 有孤立点的严格反例; [A17] 的无孤立点分量界包含等号 B , 并排除大于 B 的值。由此 $Q = 105$ 完整闭合。

6. $Q = 96$: 三个双 (14, 1) 支撑余项

size-13 已排除, 因此两侧都必须是 (14, 1)。多数块的 14 个顶点若没有孤立点, 至少需要七条内部边, 总能量至少为 $7 \cdot 9 + 42 = 105$, 超过当前的 96。因此只需考虑有内部孤立点的分支。

[P35, §10] 的双侧核支撑与平坦矩形秩约束, 在完整小分量目录中只保留三个签名。新程序重新枚举具有这些签名的图支撑、范数与相位, 确认其分别为平衡的 P_3 、 C_4 和 $K_{2,3}$, 所有圈相位均可规范化为零。这里“平衡”只用于描述核心 Gram 边; 后续原始三次单位根列测试保留全部实际边相位。

多数块支撑	孤立点数	内部能量	交叉能量	规范相位—交叉状态	$D > B$ 范数候选
$P_3 + C_4$	7	54	42	7776	52
C_4	10	36	60	4536	0
$K_{2,3}$	9	54	42	1296	0

对于交叉能量 60 的情形, 超过最小值 42 的 18 单位能量可置于核心或任意孤立点。程序保留一个平方模 21 的项、两个平方模 12 的项及各自全部余类轨道; 孤立点仅按行置换合并, 没有被删除。核心内所有六次单位相位都被作为 Gram 标签枚举。

52 个余下 Gram 全部执行两种列方程穷举, 其允许列的张成维数均至多为 9, 故不可能组成可逆的 15 列矩阵。每一个都有独立的整数 Eisenstein 湮灭向量。这完成 $Q = 96$ 的最后分支。

7. $Q = 99$: 完整纯色目录与谱迹不等式

size-13 的 67 个候选已经全部排除, 所以行、列 Gram 都必须是全同色。令

$$T = (G - 15I)/3, \quad S = (K - 15I)/3, \quad TH = HS.$$

7.1 有孤立点和无孤立点的完整覆盖

若 T 有 r 个支撑孤立点, 对应的 r 行 H 是范数平方 15 的正交向量, 且都属于 $\ker S$ 。因此

$$15(P_{\ker S})_{jj} \geq r \quad \text{对每个坐标 } j.$$

支撑叶子会迫使其邻点在全核向量中为零, 与此不等式矛盾。当前支撑最多有 11 条边; 一个没有孤立点也没有叶子的 15 顶点图至少有 15 条边。因此两侧要么都无孤立点, 要么都含孤立点, 后一种情形的非平凡分量都无叶子。

复用 [P35, §12] 中已经完整生成的 12 能量单位目录, 再限制总能量为 11。七个顶点以内的连通支撑使用完整图册; 较大支撑由全部非同构树加全部允许额外边生成, 随后精确同构去重。每个目标连通图都有生成树, 因此这一构造不会遗漏目标支撑。边范数、单位轨道与所有弦相位均完整枚举。

有孤立点的分支得到 435 个分量组合, 行列式和范数筛后保留 71 个压缩的整图谱状态。相同谱合并时取最大的最小核投影对角, 始终向保留候选的方向放宽。再要求两侧同谱并同时满足上述投影不等式, 候选全部消失。

无孤立点的分支得到 632 个分量组合。超过 B 且满足范数条件的候选有 12 个整图谱。谱迹测试需要知道各连通分量的谱, 故第二次组装保留全部分量分解, 最终得到 14 个分量—谱组合和 16 对无序的同谱行列配对。仅保留 12 个整图谱中的任意代表不够; 本次对全部 14 个分解及全部 16 个配对逐一处理。

7.2 一条无需近似特征值的精确矩形上界

谱迹引理。 设 I, J 分别是 T, S 中若干完整连通分量的顶点并集, $X = H_{I,J}$ 。记

$$p(x) = \gcd(\chi_{T_I}(x), \chi_{S_J}(x)) = x^d + c_1 x^{d-1} + \dots.$$

若 $d = 0$, 约定 $c_1 = 0$ 。则必要条件为

$$|I||J| = \|X\|_F^2 \leq 15d - 3c_1.$$

证明。 连通分量封闭性给出 $T_I X = X S_J$ 。在两个 Hermitian 矩阵的正交特征分解下, X 只能在相同特征值的空间之间非零。对于公共特征值 λ , 该块秩至多为两侧重数的较小者。此外 $XX^* \preceq G_I = 15I + 3T_I$, 故该块的每个非零奇异值平方至多为 $15 + 3\lambda$ 。对全部公共特征值按最小重数求和, 得到 $15d + 3 \sum \lambda = 15d - 3c_1$ 。另一方面, 每个 X_{ij} 模长都为 1, 故左端是矩形面积。证毕。

这一不等式对任意单位模因子已经成立, 不限于三次单位根。它保留不同特征值各自的贡献, 比“公共谱维数乘一个统一最大特征值”更强, 且右端完全由整数多项式的最大公因子给出。

程序枚举每个配对中两侧所有非空分量并集, 为全部 16 对都找到违反该不等式的矩形。此前较弱上界留下的三对, 新的整数矛盾分别为

$$40 \leq 30, \quad 20 \leq 18, \quad 36 \leq 30.$$

对应的公共多项式分别为 $x^2 - 1$ 、 $x - 1$ 、 $x^2 - 1$ 。全部 16 个证书均记录左右分量编号、两侧特征多项式、最大公因子、矩形面积与谱迹上界。由此完成 $Q = 99$ 的全同色分支和整个壳层。

8. 全局结论的合并与等号矩阵

三个新排除都针对 $D > B$ 的严格反例。size-13 不可实现性先消除了不同颜色之间的可能配对; 随后分别处理强制的双 (14, 1) 或双全同色情形。所有依赖的有孤立点分支都保留了其真实交叉约束。

最终聚合器检查三个余项与旧精确分量表之间的覆盖关系，核对 142 个 size-13 代表、52 个额外 $Q = 96$ 代表、193 个湮灭向量、 $Q = 105$ 解析矛盾以及 $Q = 99$ 全部 16 对谱证书。与历史 35 个壳层和 $Q \geq 171$ 的无限尾部联合，覆盖全部可能的 H 。

已知矩阵的精确行列式为

$$\det H_0 = 604661760 + 241864704\omega,$$

其范数就是 B 。原审计使用 Eisenstein Bareiss、有理高斯消元，以及直接计算 $\det(H_0 H_0^*)$ 三条途径核对，并确认行列能量均为 105。该矩阵的 225 个指数保留在 `order15_mu3_unified_audit/data/benchmark.json`；附录 A 也给出其逐行编码。

所以最大值等于 B ，其相对 Hadamard 界的行列式模比例约为 0.7965900126。这个比例来自已验证的矩阵与上界，不是随机搜索的经验估计。

9. 完整复演、独立验证与工程边界

在交付包根目录安装 `requirements.txt` 中的四个固定版本依赖后运行：

```
python run_final.py
```

程序默认重新执行原独立审计、前一阶段全部目录生成及证明程序，再从头生成三个最后壳层的全部新证书。`--cached-baseline` 只复用四个哈希核验过的前阶段目录；新候选、两种全量列枚举和原独立审计仍会运行。不得使用 `python -0`。

本次完整执行耗时及逐程序耗时记录在 `results/final_replay_summary.json`。新的总复演包括 13 个顶层步骤，全部退出码为零；旧 35 层在其中又完成一次全量复演。总耗时为 595.64 秒，此数值仅描述本次容器环境，不作为其他机器的运行时间承诺。

新的每种列算法均检查 $194 \times 3^{14} = 927,895,986$ 个向量赋值，两种算法合计 1,855,791,972 次。对于每个 Gram，它们输出的完整允许列序列完全一致。此外，194 个候选都接受任意精度行列式、颜色同余与逆矩阵恒等式核对，193 个湮灭向量在其对应全部允许列上逐项验证。

`provenance/final_source_sha256.json` 固定本次新增源文件；原来源清单仍单独检查。新的程序在缺文件、源哈希不符、候选覆盖不符、任意证明步骤失败或仍有余项时均报错。运行开始会把新的执行摘要置为未完成状态，因此中断不能留下一个看似属于本轮的成功摘要。原 35 层入口继续诚实地报告其历史范围。

检查结论的最短路径是先读本报告第 2、5、7 节，再核对 `last_three/final_certificate.json` 与完整日志。核验有限计算时应重新运行程序；单独查看 JSON 中的“完成”字段不能替代生成算法和数学覆盖说明。

10. 外部依赖、引文勘误与未声称事项

[TB20] 的 Proposition 14 和 Table 1 给出 $B_3(15, 10) = 12$ 。若有 13 个彼此正交的三次单位根行，将符号映射为三元码字，每对码字的 Hamming 距离为 10，会违反此界。此用途只需已发表的上界，无需重新构造其最优码。

前一份 35 层报告误写了 [TB20] 的期号和页码。正确书目信息为 *Journal of Mathematical and Computational Science* **10 (2020), No. 6, 2713–2721**, DOI 10.28919/jmcs/4964。该处引文定位现已通过出版商 PDF 核对；所用 $B_3(15, 10) = 12$ 数值没有变化。旧报告作为历史文件保留，勘误在此明确记录。

本续篇不使用旧颜色轨道方向错误所支持的孤立点排除，也不把局部候选的不可实现性自动推广到未枚举的支撑。全局范围来自第 2 节的旧约化、此次完整候选覆盖和第 8 节的合并。关于所有等号矩阵的分类、等价类数量以及形式化证明助手验证，均未包含在本次结论中。

参考文献与内部证书

[TB20] Todor Todorov and Galina Bogdanova. **Ternary equidistant codes of length** $11 \leq n \leq 15$. *Journal of Mathematical and Computational Science* 10 (2020), No. 6, pp. 2713–2721. DOI: 10.28919/jmcs/4964. 使用 Proposition 14 和 Table 1; 未复演其外部码分类。

[A17] 本项目原独立审计: `order15_mu3_unified_audit/`。提供纪录矩阵的三重精确验证、颜色及能量约化、无限尾部、最初 19 个壳层与无孤立点分量界。

[P35] 本项目《15 阶三次单位根最大行列式: 新增 16 个壳层的证明》, 2026-09-18, `report/PROOF_zh.md` 及对应 PDF。提供单边缺陷引理、双侧核投影与秩约束、扩展纯色目录, 以及前一阶段的 35/38 完整证明范围。

[F38] 本续篇: `last_three/` 的精确程序与结果, `run_final.py` 的完整复演记录, `results/final_ledger.json` 的全部 38 层状态与最终定理。

附录 A. 达到上界的矩阵

下面每一行是一个长度 15 的指数串, 数字 $e_{ij} \in \{0, 1, 2\}$ 对应矩阵项 $H_{ij} = \omega^{e_{ij}}$ 。这份数据来自原审计的纪录矩阵, 没有另行搜索或替换:

```
001000102102020
200221220112021
201110011122201
022200010001001
102122000102202
020211201100200
010021110121200
012112021111020
201212120201100
000110200221012
000122111000121
210101002101111
111212100020011
210012212002002
110010020200221
```

把每行转为 15 个指数后即可重建 H_0 。完整数据、原仓库来源与提交定位见 `data/benchmark.json`。本报告只需要一个达到上界的实例; 其他等号实例是否属于同一等价类不影响最大值定理。

历史基础证明（一）

35 / 38 阶段的原始证明报告

后附文档是 2026 年 9 月 18 日的《15 阶三次单位根最大行列式：新增 16 个壳层的证明》，共 8 页。它提供单边缺陷引理、分量 Schur 界、双侧核投影与秩约束、完整扩展纯色目录，以及当时的 35 个完整壳层证明。

文档中所述“ $Q=96, 99, 105$ 尚未闭合”属于该阶段的历史状态。这三个缺口已由本合订本第一部分的最终续篇补齐。为便于核验研究来源，历史报告内容及其原页码全部保留，不直接覆盖成新的状态。

原文引用的 Todorov-Bogdanova 论文期号、页码有误；正确书目信息和数值核对见最终续篇第 10 节。这里保留的数学依赖仍为 $B_3(15,10)=12$ 。

15 阶三次单位根最大行列式：新增 16 个壳层的证明

独立复核的继续工作·35 / 38 已闭合·3 个壳层仍待证明

研究工作记录与可重跑证书

2026 年 9 月 18 日

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1. 本次结论与适用范围

令 $\omega^2 + \omega + 1 = 0$, $\mu_3 = \{1, \omega, \omega^2\}$, $H \in \mu_3^{15 \times 15}$, 并记

$$G = HH^*, \quad K = H^*H, \quad Q = \sum_{i < j} |G_{ij}|^2, \quad B = 277868041444786176 = 2^{22}3^{20}19.$$

本包在原独立审计的基础上, 新闭合下列 **16 个完整壳层**:

$$87, 90, 108, 114, 117, 123, 126, 132, 135, 141, 144, 150, 153, 159, 162, 168.$$

合并先前已验证的 19 个壳层, 现有证明覆盖 **35 / 38**。尚未闭合的完整壳层为

$$Q \in \{96, 99, 105\}.$$

本次定理。沿用原审计中已证明的全局约化, 以及已发表的 $B_3(15, 10) = 12$ 码界, 若 H 严格超过纪录, 即 $|\det H|^2 > B$,

则必有 $Q \in \{96, 99, 105\}$ 。因此，本包没有证明全局最大值等于 B 。纪录矩阵在 $Q = 105$ 达到 B ；该事实给出下界，仍不足以处理同壳层的全部其他矩阵。

程序负责有限枚举、整数运算和有理不等式；下面给出枚举所需的结构引理与覆盖论证。它是一份计算机辅助数学证明及其复演材料，并非证明助手中的形式化推导。报告中的小数只供阅读，所有排除判据使用整数、有理数或者经精确实根隔离认证的代数信息。

2. 保留的基础与被替换的旧论证

原审计的全部 19 个已闭合壳层为 0, 6, 9, 15, 18, 24, 27, 33, 36, 42, 45, 51, 54, 60, 63, 69, 72, 78, 81。它还证明 $Q \not\equiv 0, 6 \pmod{9}$ 的壳层为空，并以迹—方差界排除全部 $Q \geq 171$ 。对于严格反例，行颜色与列颜色分别只需考虑 (15)、(14, 1)、(13, 2)、(13, 1, 1)。这里颜色由指数和模 3 决定，同色非零 Gram 项的平方模为 $9u$ ；异色项平方模至少为 3，且为 $3 \pmod{9}$ 。

对于 $Q \equiv 6 \pmod{9}$ ，可能的颜色型为 (14, 1) 或 (13, 2)；对于 $Q \equiv 0 \pmod{9}$ ，为全同色或 (13, 1, 1)。(14, 1) 的交叉能量至少为 42，size-13 两枢纽情形至少为 78。

原远端材料曾借助颜色轨道方向，删去 (14, 1) 内部支撑中的孤立点。仓库 REMOTE_AUDIT.md 已指出其中的共轭方向错误。本次不使用这个排除。下面的分量枚举明确保留全部内部孤立点；后面的交织约束直接来自 $GH = HK$ ，不假定行、列交叉项处于相同方向的颜色轨道。

外部数学依赖保留为 Todorov–Bogdanova 的 $B_3(15, 10) = 12$ ：长度 15 的三次单位根行最多有 12 个彼此正交。原审计引用 *Journal of Mathematical and Computational Science* 10(5), 1732–1739 (2020), DOI 10.28919/jmcs/4964。本次没有重新运行该论文的码分类，也不将该分类计入新枚举数量。

3. 含孤立点的统一分量—Schur 界

设多数颜色的内部 Gram 块分解成连通分量，大小与能量单位分别为 (v_i, u_i) 。每个内部孤立点单独记为 $(1, 0)$ 。连通性只要求 $u_i \geq v_i - 1$ ，从而枚举所有满足顶点总数及能量总数的无序配置，是实际图支撑的一个完整放宽。设

$$\det A_i \leq D_i, \quad \lambda_{\max}(A_i) \leq L_i.$$

对于 (14, 1)，记总交叉能量为 c 。每个分量上的交叉向量平方范数至少为 $3v_i$ 。把超过 42 的能量全部分给最大的 L_i ，只会减弱 Schur 损失。因此

$$s_{14} = \sum_i \frac{3v_i}{L_i} + \frac{c - 42}{\max_i L_i}, \quad \det G \leq \left(\prod_i D_i \right) \max\{0, 15 - s_{14}\}.$$

对于 size-13 两枢纽情形，令多数块为 A 、交叉矩阵为 U 、外部 Gram 项为 g ， $h = |g|^2$ ，交叉能量为 c 。写

$$U^* A^{-1} U = \begin{pmatrix} x & z \\ \bar{z} & y \end{pmatrix}, \quad s = x + y.$$

有 $|z| \leq \sqrt{xy} \leq s/2$ ，故

$$\det \begin{pmatrix} 15 - x & g - z \\ \bar{g} - \bar{z} & 15 - y \end{pmatrix} \leq (15 - s/2)^2 - \max\{0, |g| - s/2\}^2.$$

右端在 $0 \leq s < 30$ 上单调递减。代入

$$s_{13} = \sum_i \frac{6v_i}{L_i} + \frac{c-78}{\max_i L_i}$$

即得可逐配置计算的上界。程序使用 $\sqrt{h}$ 的有理下界，保持排除方向正确。与只用整个多数块一个谱上界相比，这里保留了孤立点和小分量各自的逆矩阵损失。

若 $(14, 1)$ 多数块有 r 个内部孤立点，则 G 至少有 $r-1$ 个特征值等于 15；两枢纽 size-13 至少有 $r-2$ 个。这些由交叉矩阵的秩直接给出，无需使用另一侧 Gram。程序同时取这一降维迹—方差界与上述 Schur 界的较小值。

4. 分量谱界、三角形数与四阶对数界

对大小 v 、能量 $9u$ 的块 $15I + E$ ，有

$$\lambda_{\max} \leq 15 + \sqrt{18u(v-1)/v}.$$

若支撑的团数至多为 k ，加权 Cauchy–Schwarz 与 Motzkin–Straus 不等式给出

$$\lambda_{\max} \leq 15 + \sqrt{18u(1-1/k)}.$$

树分量的相位可由对角酉变换消除。其行列式通过匹配多项式精确计算，原审计中 2 至 14 个顶点的树目录被保留并重跑；一般稀疏小分量则重新枚举。

为限制第三矩，先考虑连通支撑的圈秩 $g = m - v + 1$ 。三角形数的上界 $f(g)$ 可以按双连通块相加。一个非平凡双连通块的最小度 d 满足 $g \geq d(d-1)/2$ ；删除一个最小度顶点仍连通，圈秩减少 $d-1$ ，被删除的三角形至多 $\binom{d}{2}$ 。因此对较小圈秩递推，兼顾不同双连通块的分配，得到合法的上界。程序算得 $f(1), \dots, f(10) = 1, 2, 4, 5, 7, 10, 11, 13, 16, 20$ 。这些值用于上界，毋须宣称每一个值都在当前顶点数下达到。

设边平方模为 $9w_e$ ，加权三角和为 $W = \sum_{\triangle} \sqrt{w_1 w_2 w_3}$ 。枚举有限范数分拆后，可以用三角形数及最大的三个权重放宽 W 。程序还使用 $\sqrt{w_e} = 1 + \delta_e$ 的展开：一条边至多属于 g 个三角形，一对不同边至多同属一个三角形，从而给出另一条更紧的加权上界。所有根式均向上取有理括号。

以下四阶界把这些离散限制转化为行列式损失。对 $x > -1$ 且 $x \leq M$ ，积分余项给出

$$\log(1+x) \leq x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4(1+M)}.$$

取 $M = L/15 - 1$ ，其中 L 是块的最大特征值上界。若 $\text{rank } E \leq d$ ，则

$$\text{tr } E^2 = 18u, \quad \text{tr } E^3 \leq 162W, \quad \text{tr } E^4 \geq \frac{(18u)^2}{d}.$$

求和得到

$$\det(15I + E) \leq 15^v \exp \left(-\frac{u}{25} + \frac{2W}{125} - \frac{3u^2}{125Ld} \right).$$

若括号的相反数为 $a \geq 0$ ，程序用 $(\sum_{j=0}^{24} a^j/j!)^{-1} \geq e^{-a}$ 给出纯有理上界。四阶项在高能量、低秩的余项中非常重要；单独的三阶估计不足以完成同样的排除。

5. 直接完成的六个混色情形

`explore.py` 对全部允许的内部能量、外部项范数与分量配置使用上述界，包括内部能量为零及所有孤立点数。下表的两个数是各颜色分支上界与 B 的比值；精确分子、分母由复演程序产生，小数为阅读近似。

Q	size-13 分支最大上界 / B	(14, 1) 分支最大上界 / B
123	0.965876	0.984599
132	0.943734	0.979425
141	0.922622	0.971827
150	0.896802	0.963220
159	0.877307	0.939258
168	0.856213	0.919227

每一行覆盖该壳层的所有允许颜色，因此是完整壳层闭合。其中 $Q = 159$ 的有孤立点分支已被纳入；旧论证中遗漏该分支的问题没有被继承。

6. 全同色的双侧核投影约束

在全同色情形，写 $T = (G - 15I)/3$, $S = (K - 15I)/3$ 。有 $TH = HS$ 。若 T 有 r 个完整支撑孤立点，相应的 r 行 H 彼此正交，且都属于 $\ker S$ ，因此

$$15(P_{\ker S})_{jj} \geq r \quad \text{对每个 } j.$$

这迫使另一侧每个坐标的核投影均为正。支撑中的叶子会迫使其邻点在全部核向量中为零，因而不能出现。在当前 $u = Q/9 < 15$ 的范围，另一侧若没有孤立点又没有叶子，就至少需要 15 条边，矛盾。因此两侧要么均无孤立点，要么均有孤立点；后一情形两侧均无叶子。

若另一侧的孤立点数至少为 r_0 ，一个大小 v 的连通分量就必须满足

$$\dim \ker T_i = \text{tr } P_{\ker T_i} \geq \left\lceil \frac{vr_0}{15} \right\rceil.$$

对当前候选集取最小的 r_0 ，利用这个秩限制重新计算分量界，删除被排除者，再更新 r_0 ，得到安全的单调筛选。它允许不同支撑、相位与两侧图形，无须把行 Gram 与列 Gram 预先认作同一轨道。

`projection_frontier.py` 由此直接排除 $Q = 135, 144, 153, 162$ 的全部全同色候选；它们的 size-13 分支已经由第 3 节的统一界排除，故四层全部闭合。 $Q = 117, 126$ 进一步约化到下面的低秩整数问题。

7. $Q = 117, 126$ 的整数特征多项式证书

因为 T 是 Eisenstein 整数 Hermitian 矩阵，其特征多项式系数同时属于 $\mathbb{Z}[\omega]$ 与实数，故属于 $\mathbb{Z}$ 。若 $\text{rank } T \leq d$ ，则

$$\det G = 15^{15-d} 3^d R, \quad R \in \mathbb{Z}.$$

这里特征值 15 的重数给出粗迹界之外很强的整数网格。每个候选 $\det G$ 还必须是 Eisenstein 范数：分解到有理素数后，所有 $p \equiv 2 \pmod{3}$ 的指数均为偶数。

对于 $Q = 126$, 仅剩一个 8 顶点、14 能量单位的连通分量和 7 个孤立点, 核投影迫使其秩至多为 4。有理上界与 $D > B$ 共同把 R 限制为 397, 398, 399, 400, 401。五个整数 $15^{11}3^4R$ 全部不是 Eisenstein 范数, 故该层闭合。这里检验的是带前因子的完整行列式, 不能把检验误写为只检查 R 本身。

对于 $Q = 117$, 投影后只剩大小 7、8 的秩至多 4 情形, 以及大小 9 的秩至多 5 情形。秩 4 的范数筛只留下 $R = 405$ 。约化特征多项式具有形式

$$p(x) = x^4 - 13x^2 - tx + 105 - 5t, \quad -22 \leq t \leq 22.$$

程序以精确实根隔离检查全部 45 个多项式, 没有一个具有所需的四个实根。秩 5 的四阶界和范数筛把 R 限制为 1983, 1984, 1987; 多项式具有形式

$$p(x) = x^5 - 13x^3 - tx^2 + e_4x - e_5, \quad R = 1500 + 25t + 5e_4 + e_5.$$

三角和限制 $|t| \leq 12$, 第四矩与绝对值 AM-GM 分别给出安全范围 $-85 \leq e_4 \leq 50$ 、 $|e_5| \leq 62$ 。代入整数关系后共有 240 个待检多项式, 也全部未通过实根要求。两种秩情形均排除, 完成 $Q = 117$ 。

8. $Q = 114$: 最后一个抽象 Gram 也不能分解为平坦矩阵

统一界已排除整个 size-13 分支。(14, 1) 只剩 8 个连通分量配置。逐个枚举内部范数、支撑与树规范化后的弦相位, 再精确枚举交叉向量二次型 $z^*A^{-1}z$, 与外部孤立点和小树的损失相加。超过 B 的值逐一接受 Eisenstein 范数检验。

最后只剩 K_4 加 10 个内部孤立点的一个抽象 Gram 轨道, 其行列式为

$$D_* = 278010073125000000 \approx 1.000511148B.$$

注意 $D_* > B$; 故到这一刻还不能宣称闭合。最后分支的 3024 个交叉向量状态中, 达到必要最小损失的 6 个状态均可规范化成多数块 $12I_4 + 3J_4$, 核心交叉向量为 $\delta(2, 2, 1, 1)$, $\delta = 1 - \omega$, 其余 10 个交叉项为 δ 。

精确求核表明, G 的 12-特征空间只支撑于这四个核心坐标, 投影对角为 $(1/2, 1/2, 1/2, 1/2, 0, \dots, 0)$, 具体是两组等交叉权重坐标的差方向。两侧 Gram 若来自 H , 必须都处于这个唯一余项轨道。 $GH = HK$ 迫使相应的两列在核心之外的 11 个位置成比例, 且各元素模长为 1。因此该两列内积的模至少为 $11 - 4 = 7$, 但规定的 Gram 内积模为 3, 矛盾。这个最后排除对任意单位模矩阵已经成立, 不依赖有争议的颜色方向。

9. 新的单边缺陷引理: 13 行中一个 norm-9 内积不可能

引理。不存在 13 个长度 15 的三次单位根行, 其 Gram 非对角支撑恰好是一条平方模为 9 的边。

取其中 12 个彼此正交的行组成 R , 把与剩余行 h 有非零内积的那一行规范化为全 1。再用允许的三次单位根变换使 $\langle h, \mathbf{1} \rangle = 3s$, $s \in \{1, -1\}$ 。令

$$M = 15I - R^*R.$$

则 $M \succeq 0$ 、秩为 3、对角为 3、 $M^2 = 15M$ 。非对角项为 Eisenstein 整数, 其平方模只能是 0, 3, 9。把同一复直线上的列归为一类, 代表单位向量为 $v_i \in \mathbb{C}^3$, 重数为正整数 m_i ; 平行列的相位比属于六次单位根。不同直线间的平方内积只能为 0 或 $1/3$ 。

令 O 为这些直线的正交图。由 $\sum m_i v_i v_i^* = 5I$ 得到 $Om = 2m$ 。对任一连通分量 C , 令 $M_C = \sum_{i \in C} m_i$ 。直接计算其加权框架算子的迹与平方迹, 得到

$$\sum_{i \in C} m_i v_i v_i^* = (M_C/3)I.$$

直线投影矩阵的 Gram 为 $(2I - O_C + J)/3$ 。Perron–Frobenius 给出 $2I - O_C \succeq 0$ ，其核由正向量生成；加上 J 后正定。它嵌入 3 阶 Hermitian 矩阵的 9 维实空间，因此每个连通分量至多有 9 个直线顶点。

接下来给出需要的谱半径 2 图分类。含圈者只能是圈本身，否则严格增大 Perron 根；有度数至少 4 者只能是四叶星。其余为度数至多 3 的树。含两个分叉点者必须是两分叉之间的一条路径，两端各接两片叶，因为这个子图已经有谱半径 2。仅一个分叉点者的三臂长度 (a, b, c) 满足 $1/(a+1) + 1/(b+1) + 1/(c+1) = 1$ ，从而只有 $(2, 2, 2), (1, 3, 3), (1, 2, 5)$ 。这完成所需的有限分类，无需把仿射图分类当作未经证明的黑盒。

取原始正整数权重 $m^{(0)}$ 。直线代表的 Gram N 满足 $N \operatorname{diag}(m^{(0)})N = M^{(0)}N$ 。左侧各项被理想 (3) 整除。除正交三角形外，存在平方模为 3 的非对角项，故 $3 \mid M^{(0)}$ ；正交三角形的原始质量本身就是 3。结合至多 9 个顶点、总质量不超过 15，允许的连通质量只有 3, 6, 9, 12, 15，且质量 15 只能来自三个正交方向、每个重数为 5。

最后令 $z = h - (s/5)\mathbf{1}$ 。它属于秩 3 补空间。坐标 $5z_j = 5h_j - s$ 的三种可能值彼此都不是六次单位根倍数，所以每一个平行列类别只能包含同一种 h_j 。各连通分量的紧框架等式又迫使 $|5z_j|^2$ 的加权平均为 24。 $s = 1$ 时三个平方模为 16, 31, 31，其中第一种坐标数必须为 $7M_C/15$ ； $s = -1$ 时为 36, 21, 21，第一种必须为 $M_C/5$ 。因此每个连通质量被 5 整除，结合前面的 3 整除，只能是质量 15 的正交三角形，各符号数均为 5 的倍数。但所需的全 1 符号数分别为 7 或 3，矛盾。

`sparse_defect.py` 精确检查图的正特征向量、允许质量、残余坐标平方模与六次单位根轨道；上述各段提供这些有限检查的结构依据。此引理排除了所有 size-13 内部能量 $e = 9$ 分支。

10. 正确的 $(14, 1)$ 双侧秩约束与 $Q = 87$

设行侧多数颜色内部有 r 个孤立点，非孤立内部矩阵 T_A 的核维数为 k ，其核支撑坐标数为 t 。另一侧相应记为 R, K_0, T_0 。由于孤立点仍与单枢纽相连，整个 $G - 15I$ 的核维数为 $r - 1 + k$ ，且核支撑至多有 $r + t$ 个坐标，枢纽坐标始终为零。对两侧内部孤立点 i, j ，从 $(G - 15I)H = H(K - 15I)$ 的 (i, j) 项可得交叉模长相等。把孤立行除以其交叉项后，任意两行之差属于另一侧的核。这给出三个必要条件及其反向版本：

$$r + k = R + K_0, \quad r(15 - R - T_0) \leq 15, \quad r(15 - R) \leq 15(K_0 + 1).$$

第二条来自另一侧核支撑以外的相同比例列贡献：这些列在 r 个正交行上形成秩一矩阵。第三条保留另一侧非孤立内部核的维数：由孤立行限制到另一侧非孤立内部块及枢纽得到的矩形，秩至多为 $K_0 + 1$ ；其平方 Frobenius 范数为 $r(15 - R)$ ，而最大奇异值平方至多为 15。上述推理直接使用交织与正交性，未比较错误方向的颜色轨道。

`rank_catalogue.py` 完全枚举所需的能量至多 7 单位的连通分量，记录核维数、核支撑、独立数与行列式上界。 $Q = 87$ 经码界与纪录比较后仅有 44 个抽象组合；应用双侧配对条件后全部消失。其 size-13 只可能剩 $e = 0$ 或 $e = 9$ ，分别由已发表码界和第 9 节引理排除，故 $Q = 87$ 完整闭合。

同一计算也排除 $Q = 105$ 的严格双 $(14, 1)$ 有孤立点分支；无孤立点等号分支由原审计处理。但另一个颜色型尚未全部排除，所以这里不把 $Q = 105$ 标为闭合。

11. $Q = 90$ 的完全目录与谱矩形矛盾

size-13 分支已由 $e = 0, 9$ 两个结论排除，因而两侧均为全同色。两侧有无孤立点的二分由第 6 节成立。无孤立点时，10 个能量单位的预算迫使每个连通分量至多 7 个顶点；有孤立点时，没有叶子的连通分量至多 10 个顶点。

7 顶点以内使用完整图册。较大的无叶支撑由所有非同构生成树加上所有允许的额外边得到，再以精确同构检验去重。任一目标图都有生成树，因此此生成方式覆盖全部目标图；最小度与能量限制只删去已证明不需要的支撑。每条树边按六次单位根规范化，每条弦穷举六种相位，并枚举全部允许范数及不同范数轨道。

对每个归一化矩阵 T_i 精确计算特征多项式。若 $p(x) = x^k q(x)$ 、 $q(0) \neq 0$ ，则 $P_{\ker T_i} = q(T_i)/q(0)$ ，故核投影对角也用整数矩阵多项式精确求出。相同谱的状态合并时保留最大的最小投影对角，使筛选始终偏向保留候选。

有孤立点的候选全部在范数与双侧投影筛中被排除。无孤立点者只剩两种多项式，结构均为五个 K_2 和一个 5 顶点分量：

$$p_T(x) = (x^2 - 1)^5 f_i(x),$$

$$f_1 = x^5 - 5x^3 + 5x - 1, \quad f_2 = x^5 - 5x^3 + 2x^2 + 4x - 2.$$

两侧 Gram 的谱相同。5 顶点块与对侧 10 个 K_2 坐标之间的矩形 X 满足相应 Sylvester 方程。 f_1 与 $(x^2 - 1)^5$ 的最大公因子次数为 1， f_2 的为 2，所以 $\text{rank } X \leq 2$ 。该矩形有 50 个单位模元素，故 $\|X\|_F^2 = 50$ 。另一方面，5 顶点块的交叉能量为 45，给出 $\lambda_{\max} \leq 15 + \sqrt{72} < 24$ ，而 $XX^* \preceq G_5$ ，从而 $50 < 48$ ，矛盾。两个谱候选全部排除，完成 $Q = 90$ 。

12. $Q = 108$ ：完整扩展目录与 size-13 能量 18

这是本次最后完成的一层。统一界与单边缺陷引理之后，size-13 只剩内部能量 18、交叉能量 78、外部平方模 12。内部支撑只能是 P_3 加 10 个孤立点，或 $2K_2$ 加 9 个孤立点。

这里使用同一个 Gram 内部的正确颜色信息。两根交叉列都处于最小范数 3，在每一行上的比值属于同一个三次单位根陪集。对一个枢纽作整体六次单位根变换后，可以写每行的交叉项为 $(z_i, z_i t_i)$ ， $|z_i|^2 = 3$ 、 $t_i \in \mu_3$ 。这个比值在内部树规范化时保持不变；它没有涉及行、列 Gram 的方向比较。

对 P_3 和两个 K_2 ，逐一枚举核心中的 z_i 与 t_i ，精确记录

$$x = u^* A^{-1} u, \quad y = v^* A^{-1} v, \quad z = u^* A^{-1} v.$$

r 个内部孤立点各贡献 $1/5$ 到两个对角损失，非对角只通过 $s = n_0 + n_1 \omega + n_2 \omega^2$ 出现，其中 $n_0 + n_1 + n_2 = r$ 。因此可以精确枚举有限的三个符号计数，完全保留相位约束，而不需要枚举每个坐标的全部排列。最终比较量为

$$\det A \, 15^r [(15 - x - r/5)(15 - y - r/5) - |g - z - s/5|^2].$$

两个支撑在 $|g|^2 = 12$ 时的全部值都低于 B ；这一步甚至不需要范数筛。validate_size13.py 另外选取 16 个完整 15 阶实例，以独立的任意精度 Bareiss 行列式核对这一压缩公式。

全同色部分采用扩展到 12 能量单位的完整目录。对于有叶子的分量，若要置于无孤立点的 15 顶点总图中，必须满足 $u + \lceil (15 - v)/2 \rceil \leq 12$ ；其余支撑只需保留最小度至少 2 的情形。大于 7 个顶点的必要图仍按生成树加全体额外边枚举，最大顶点数为 12。目录共实际计算 15,899,506 个相位状态，保留 4,555 个压缩谱类型。

无孤立点情形共检查 5,570 个组装，没有任何超过 B 且满足范数条件的候选；有孤立点情形共检查 146 个组装，范数筛后剩 8 个，随后全部被双侧核投影条件排除。因此 $Q = 108$ 的两类颜色均被排除，完整闭合。

13. 程序覆盖、算术可靠性与实际复演

四个主要有限目录的实际计数如下。它们的作用不同，不能把相位状态数量直接解释成不同原始 H 的数量。

目录	实际相位状态数	保存的对象
小分量行列式目录	514,858	45 个 (v, u) 最大值记录
小能量秩与核支撑目录	2,914	92 个签名
$Q = 90$ 纯色目录	531,268	29,402 个谱类型
$Q = 108$ 扩展纯色目录	15,899,506	4,555 个筛后谱类型

加速内核使用有符号 64 位整数，未把浮点近似用作证明判断。归一化矩阵为 $5I + T$ ，所有被枚举的 $\sum_{i < j} |T_{ij}|^2 \leq 12$ ，故 $\|T\|_2 \leq \sqrt{24} < 5$ ，矩阵正定。Hadamard-Fischer 对所有子式的界，以及最多 12 阶的维数，保证 Bareiss 更新中间整数远小于 2^{63} 。特征多项式与核投影的矩阵幂也受 $\|T\| < 5$ 和二项式系数界控制。算法在每步检查实主元、整数整除与 Newton 恒等式整除条件。

`validate_primitives.py` 用独立的任意精度 Eisenstein Bareiss、Gaussian 消元、特征多项式和核投影程序核对 55 个确定性随机矩阵，阶数覆盖 2 至 12；全部通过。多重集排列生成器另外与直接排列集合核对。另有前述 16 个完整 15 阶 Schur 公式检查。随机交叉检查服务于实现核验；枚举完整性仍由本报告给出的确定性支撑、范数、相位及组装覆盖证明保证。

本包附带源码哈希、目录哈希、逐步骤标准输出和错误输出。原审计八项基础复演也在本次交付目录中重新运行；其历史结论保持 19 层闭合。新的状态表单独生成，避免覆盖原审计历史或把旧 CLOSED 字样当作新证明。

14. 剩余三层与严格的交接边界

$Q = 96, 99, 105$ 的 size-13 剩余支撑已集中为内部能量 18 的 P_3 或 $2K_2$ ，交叉能量为 78，外部平方模分别为 0、3、9。对于 $Q = 105$ ， P_3 支撑仍保留两个范数相容值 278010073125000000、278757412031250000； $2K_2$ 支撑仍保留 278010073125000000、281238577200000000。结合双 $(14, 1)$ 严格反例分支的排除，任何 $Q = 105$ 的严格反例，其行列式平方必在这三个不同整数之中。它们均尚未实现为三次单位根矩阵，也未被全部排除。先前过程消息所称“只剩一个数值”过强，应以这里的复演结果为准。全部候选值保存在 `new/size13_energy18.json`，它们只是必要条件下的抽象 Gram 候选，不能当成已构造的纪录反例。

$Q = 96$ 的双 $(14, 1)$ 内部孤立点配对还有 3 类签名，需要进一步处理或与其他颜色型一起完成谱配对。 $Q = 105$ 的双 $(14, 1)$ 严格反例分支已经排除，但 size-13 及跨颜色配对仍有缺口。 $Q = 99$ 还需要把所有可能的全同色支撑约化到完整终端目录；原审计中 A/B、A/A、B/B 的局部交织枚举可以继续使用，其适用范围不能自行扩大。

因此，本包的研究结论是 **新增 16 层完整证明，剩余 3 层**。将这三层进一步标为闭合，需要新的完整排除；修改状态字段、仅复演旧测试，或只展示某个局部 UNSAT 都不够。

15. 复演入口与文件组织

在包根目录安装 `requirements.txt` 后执行：

```
python run_progress.py --include-baseline --audit-only
```

该入口重跑原审计与全部新目录及后续证书，写出 `results/progress_ledger.json`、`results/all_Q_progress.csv` 和逐步骤日志。使用 `--cached --audit-only` 会检查交付目录的哈希并重新计算下游证书，但不会重新生成全部相位目录，输出摘要会明确记录这一点。

去掉 `--audit-only` 后，只要尚有未闭合壳层，程序以退出码 2 结束。这是对全局证明状态的明确报告，不应被改成无条件成功。`new/` 是新增可重跑源码，`order15_mu3_unified_audit/` 保留原基础，`provenance/` 保存来源、原始 ZIP 和哈希。所有新结果均在本地交付目录中；本次没有向远端 GitHub 推送修改。

历史基础证明（二）

原独立审计与最初 19 个壳层

后附文档是项目原始独立审计报告，共 13 页。它保留了原有关闭声明的审查、纪录矩阵的三条精确计算路径、能量及颜色约化、最初 19 个壳层的证明、无孤立点分量界，以及全体 $Q \geq 171$ 的尾部控制。

其中“全局最优性尚未证明”和“19 个壳层待验证”等状态描述同样属于历史阶段。第二部分新增 16 层，第一部分最终补齐最后三层；当前完整结论以本合订本第一部分以及交付包中的 `final_ledger.json` 为准。

本合订本以原文附录方式保留证据链。三个部分的原页码分别保留，使用 PDF 书签或物理页码可以在最终续篇、35 层阶段与原独立审计之间定位。

Order-15 maximal determinant over third roots of unity

Independent audit and reconstructed certificates

17 September 2026

中文结论：全局最优性尚未验证

本次复核保留原始压缩包的内容，恢复纪录矩阵，并重新编写、实际运行了一组精确验证程序。已完成全部 $Q \leq 81$ 的允许壳层，共 19 个；全部 $Q \geq 171$ 由迹—方差界排除， $Q \not\equiv 0, 6 \pmod{9}$ 由同余排除。部分低能量证明使用已发表的三元等距码定理；该外部分类没有在本次重新运行。

此外， $Q = 99$ 的三个局部交织问题已独立恢复全部解并验证矛盾； $Q = 105, 114, 123, 132, 141, 150, 159, 168$ 的 $(14, 1)$ 型内部无孤立顶点分支，全部 36 种内部能量分配均已取得纪录级上界。这些是完整的局部分支结果，仍不能替代每个壳层的完整证明。

纪录矩阵的三条精确计算路径一致给出

$$|\det H_0|^2 = B = 277868041444786176 = 2^{22} 3^{20} \cdot 19, \quad Q(H_0) = 105.$$

它给出确定的下界。剩余 19 个允许壳层仍有未完成的证明义务；原 **ledger** 中的 **CLOSED** 已与本次证据严格区分。因此，这份文件不得作为“全局最大值已精确确定”的证明提交。

1 Statement and scope of the audit

Let $\omega^2 + \omega + 1 = 0$, let $\mu_3 = \{1, \omega, \omega^2\}$, and put

$$D_3(15) = \max_{H \in \mu_3^{15 \times 15}} |\det H|^2, \quad G = HH^*, \quad K = H^*H, \quad Q = \sum_{i < j} |G_{ij}|^2.$$

Both Gram matrices have diagonal 15 and the same energy Q . A strict counterexample to the record means $\det G > B$. This terminology concerns an unproved upper bound; it does not assert that any such matrix exists.

The new computations establish that no strict counterexample has $Q \leq 81$ or $Q \geq 171$. The assertion for $Q \leq 81$ uses the external theorem $B_3(15, 10) = 12$ in the cases indicated below. Independently of that external code theorem, the arithmetic/color reductions and the nine shells $Q = 0, 9, \dots, 72$ are checked by the supplied exact computations. The four shells 6, 15, 24, 33 are also excluded directly by the color/energy reduction. The remaining six low shells 42, 51, 60, 69, 78, 81 use the cited code theorem.

A second result covers every $(14, 1)$ color partition with no internal support isolates at $Q = 105, 114, 123, 132, 141, 150, 159, 168$. Here the upper bound is at most B , with equality possible in the friendship-support case at $Q = 105$. A third result exhaustively verifies the specified A/B local obstructions at $Q = 99$. Neither result closes all other color and support types at those energies.

The global statement $D_3(15) = B$ remains *unverified by this delivery*. The unresolved energies are

$$87, 90, 96, 99, 105, 108, 114, 117, 123, 126, \\ 132, 135, 141, 144, 150, 153, 159, 162, 168.$$

The package contains the original sixteen archive files, new reconstructions, and machine-readable results. It does not contain a complete checkout of the original working repository. In particular, historical tail closures have not been promoted to verified global statements without replaying their upstream reductions.

2 Exact arithmetic and the benchmark

2.1 Arithmetic representation and cross-checks

An Eisenstein number $a + b\omega$ is represented by (a, b) , with rational coordinates when division is required. Multiplication, conjugation and norm are

$$(a, b)(c, d) = (ac - bd, ad + bc - bd), \quad \overline{(a, b)} = (a - b, -b), \quad N(a, b) = a^2 - ab + b^2.$$

All comparisons deciding a certificate are integer or rational comparisons. Floating-point values in JSON and this report are display-only approximations.

The implementation includes fraction-free Bareiss elimination and a separate rational Gaussian elimination. Sixty deterministically generated matrices of orders 1 through 6 are checked against both methods and against an independent real-integer representation: multiplication by $a + b\omega$ acts through

$$\begin{pmatrix} a & -b \\ b & a - b \end{pmatrix}, \quad \det \mathcal{R}(A) = N(\det A).$$

Exact inverse identities and characteristic-polynomial evaluations are also checked. Tree determinant dynamic programming is compared directly with Bareiss elimination for all nonisomorphic trees on 2 through 9 vertices. These tests are in `programs/verify_arithmetic_tests.py`; their counts and outcomes are recorded separately from the mathematical shell claims.

2.2 Recovered witness

The exponent matrix in `data/benchmark.json` was recovered from the repository benchmark data at commit `007af7d8f7f644396aee27dd87c68c8f4169efd5`. The original data blob is identified in the provenance metadata. The local JSON adds source information and is not claimed to be byte-identical to that blob.

The three exact paths give

$$\det H_0 = 604661760 + 241864704\omega, \quad \det(H_0 H_0^*) = N(\det H_0) = B.$$

The row and column energies are both 105. The row support degrees are fourteen occurrences of 2 and one occurrence of 14, giving the friendship support. Consequently

$$\sqrt{B} = 2^{11}3^{10}\sqrt{19}, \quad \frac{\sqrt{B}}{15^{15/2}} = 0.7965900126038639 \dots$$

This is a verified witness and hence a lower bound on $D_3(15)$. The equality with the unknown global optimum is not inferred from it.

3 Global reductions

3.1 Congruence and norm conditions

For two rows let $n_0 + n_1 + n_2 = 15$ count the three exponent differences. Their inner product is

$$z = (n_0 - n_2) + (n_1 - n_2)\omega, \quad \delta = n_1 + 2n_2 \pmod{3}.$$

Writing $a = n_0 - n_2$ and $b = n_1 - n_2$ gives $a + b = 15 - 3n_2$ and $b \equiv \delta \pmod{3}$. Thus

$$N(z) = (a + b)^2 - 3ab \equiv 3\delta^2 \pmod{9}.$$

The program also checks this identity on all 136 count triples. The row color is the sum of its exponents modulo 3. Same-color edges have squared norm divisible by 9; different-color edges have squared norm 3 modulo 9 and are nonzero. For color-class sizes $a + b + c = 15$,

$$Q \equiv 3(ab + ac + bc) \equiv 0 \text{ or } 6 \pmod{9}.$$

There are exactly 38 eligible energies between 0 and 168.

Subtracting the first row from each other row shows that $(1 - \omega)^{14}$ divides $\det H$. Therefore 3^{14} divides $\det G$. Also, any prime $p \equiv 2 \pmod{3}$ occurs to an even exponent in the nonzero norm $\det G = N(\det H)$. This latter necessary condition is used as an exact factorization sieve, never as evidence of realizability.

3.2 Trace–variance envelope

For a positive semidefinite $d \times d$ matrix of trace $15d$ and off-diagonal energy q , its eigenvalues have mean 15 and squared deviations summing to $2q$. An interior maximum of their product has at most two distinct eigenvalues, by Lagrange multipliers. Boundary products vanish. For multiplicity m the stationary product is

$$P_{d,m}(q) = (15 + (d - m)t)^m (15 - mt)^{d-m}, \quad t = \sqrt{\frac{2q}{dm(d - m)}}.$$

On the positive branch this product decreases with t . Replacing t by the certified rational lower bound

$$t_- = 10^{-15} \left\lfloor 10^{15} \sqrt{\frac{2q}{dm(d - m)}} \right\rfloor$$

therefore gives an upper bound. Taking the maximum over m yields the rational envelope $U_d(q)$ used in the code. Each branch decreases as q increases, so the maximum is nonincreasing as well.

The exact computation verifies

$$U_{15}(171) < B, \quad U_{15}(171)/B \approx 0.9944244061.$$

This excludes the entire range $Q \geq 171$, not just the single value 171. Exact numerators and denominators are retained in `results/foundations.json`.

3.3 Color partitions

If a color class of size a is internally orthogonal, the minimum cross energy is $3a(15 - a)$. A Schur complement and AM–GM give

$$\det G \leq 15^a \left(\frac{75 - a}{5} \right)^{15-a}.$$

For every $3 \leq a \leq 12$, the right-hand side is strictly below B . Hence each such class of a strict counterexample contributes at least one internal norm-9 edge. Combining this with $Q \leq 168$ leaves ten sorted partitions. Further Schur bounds exclude largest classes of sizes 10, 11 and 12 and leave only

$$(15, 0, 0), \quad (14, 1, 0), \quad (13, 2, 0), \quad (13, 1, 1).$$

For an active class of size d , internal energy e , largest-eigenvalue cap L , and cross energy at least c , the bound is

$$U_d(e) \left(15 - \frac{c}{L(15 - d)} \right)^{15-d}.$$

The verified (d, e, L, c) values are $(10, 9, 18, 150)$; for $d = 11$, $e = 9, 18, 27$ with $L = 20, 21, 23$ and $c = 132$; and for $d = 12$, $e = 9, 18, 27, 36, 45, 54$ with $L = 18, 77/4, 21, 22, 241/10, 25$ and $c = 108$. The low-energy caps follow from the finite norm catalogue and small support graphs; the larger ones follow from $\lambda_{\max}(E)^2 \leq 2e(d - 1)/d$. Small unweighted graph caps are checked exactly, including the four-disjoint-edge case beyond the seven-vertex atlas.

4 Reconstructed low-energy proofs

4.1 Isolates and the two-sided projection condition

Put $E = G - 15I$ and $F = K - 15I$. The identity $EH = HF$ is central. If F has an isolated support vertex, the corresponding column of H is a full-support vector in $\ker E$. A leaf of the support of E would force one coordinate of that vector to vanish. Therefore an isolate on one side forbids a leaf on the other.

If F has r support isolates, the corresponding columns form a matrix Y with $Y^*Y = 15I_r$ and range in $\ker E$. Hence

$$\frac{1}{15}YY^* \preceq P_{\ker E}, \quad \min_i (P_{\ker E})_{ii} \geq \frac{r}{15}. \quad (1)$$

This is a two-sided necessary condition because E and F have the same characteristic polynomial. It is stronger than testing singularity of the active support alone.

4.2 The nine shells $Q = 0, 9, \dots, 72$

For these energies the color reduction forces both Grams to be all-same-color. Up to $Q = 63$, at most seven edges cannot cover 15 vertices, so both supports have isolates. At $Q = 72$, a support without isolates must be $P_3 \sqcup 6K_2$ with norm-9 edges. Its Gram determinant is $3105 \cdot 216^6$, which fails the norm sieve: the primes 5 and 23 have odd valuations. Thus both supports again have isolates, and all active components have minimum degree at least two.

Set $k = Q/9 \leq 8$ and divide E by 3. A connected active component has at most eight edges and eight vertices. The seven-vertex graph atlas plus C_8 is exhaustive in this range: an eight-vertex minimum-degree-two graph with at most eight edges is a cycle. Edge norm units are drawn from the exact catalogue; under this budget only 1, 3 and 4 can occur. Sixth-root diagonal switching fixes spanning-tree phases, and every remaining associate class and cycle phase is enumerated. This switching is used only for spectral and projection necessary conditions, which it preserves.

For each component the program computes the characteristic polynomial and the exact kernel projection. Components with a zero projection diagonal are discarded. Component assemblies are filtered by determinant, divisibility and norm. Assemblies with a common full characteristic polynomial are then coupled by (1), iterating until no further state can be removed. When states with identical spectral data are merged, the larger projection minimum is retained; this only relaxes the necessary condition. Every remaining set is empty. No missing historical support catalogue or external equidistant-code theorem is used for these nine shells.

4.3 The small mixed shells

An orthogonal set of cube-root rows of length 15 gives a ternary equidistant code of distance 10. The published theorem $B_3(15, 10) = 12$ [1] therefore forbids 13 mutually orthogonal rows. This external classification is not rerun here.

A color class of size a with m internal support edges has an independent set of size at least $a - m$. The code theorem forces at least $a - 12$ edges when $a > 12$. Together with the color congruence and cross energy, this closes $Q = 42, 51$; $Q = 6, 15, 24, 33$ already fail the minimum mixed cross energy of 42. At $Q = 60, 69, 78$, both sides must have color type $(14, 1)$.

Let C be the nonisolated part of the internal 14-row support, and let I be its set of k isolates. Here $k \geq 6$. The full 15-eigenspace has zero hub coordinate and coordinate support contained in $I \cup \text{supp ker } C$. For the opposite Gram, the hub coordinate is likewise absent. Intertwining then forces the cross entries on I into one cube-root orbit: a column of H at that opposite hub is orthogonal to the hyperplane $u_I^\perp$ supported on I , and so is proportional to u_I . All entries of that column have modulus one and belong to μ_3 .

There are at most two excess cross-energy units, fewer than k . The common squared norm of these isolated cross entries must consequently be 3. After cube-root normalization the isolated cross entries are identical. If the opposite 15-eigenspace is supported on at most s coordinates, at least $15 - s$ columns are identical on I up to a scalar cube root. Positivity of the residual Gram matrix gives

$$15I_k - (15 - s)J_k \succeq 0, \quad k(15 - s) \leq 15. \quad (2)$$

The complete finite enumeration contains all internal supports with two to four energy units, every permitted phase, and every active cross allocation. It uses $s \leq k + |\text{supp ker } C|$, a safe upper estimate. The actual enumerated state counts are 36, 1008 and 25416 for $Q = 60, 69, 78$, respectively. After the exact norm sieve and spectral deduplication, (2) removes all states. The output explicitly labels the external code-theorem dependency.

5 The reconstructed $Q = 81$ boundary

The code theorem removes the mixed $(13, 1, 1)$ alternative, so both Grams are all-same-color. A no-isolate support has exactly nine norm-9 edges: a heavier edge would reduce the edge count to at most seven, insufficient to cover 15 vertices. Such a graph is either a forest with six components, each of size at most five, or $K_3 \sqcup 6K_2$. All forest spectra fail the required determinant/norm filters. The surviving no-isolate spectra are the four real-cycle-product possibilities for the triangle, and none has eigenvalue 15.

Therefore, if either support has an isolate, both do. The preceding kernel-projection method extends to nine energy units. For completeness beyond the seven-vertex atlas, an eight-vertex nine-edge minimum-degree-two connected graph has degree excess two. It is a figure-eight, a barbell, or a theta graph. The program explicitly generates all possibilities, together with C_8 and C_9 ; it also includes both associate classes of norm-7 edges. The three norm-surviving isolate states are eliminated by (1).

It remains to exclude $K_3 \sqcup 6K_2$. Every pair block has spectrum $\{18, 12\}$. When the triangle cycle product has real part $\pm 1/2$ after normalization, its characteristic polynomial is coprime to the pair polynomial. The corresponding 3×2 block of H would vanish by intertwining, which is impossible. Positive real cycle product gives a triangle eigenvalue 21, supported on three coordinates on each side. The 3×3 block of H would then need squared operator norm at least 21, exceeding its Frobenius norm squared 9.

For negative real cycle product, the triangle spectrum is $\{18, 18, 9\}$. The localized eigenvalue 9 saturates the same 3×3 Frobenius bound. Its block must have rank one. Since its entries are cube roots, row and column cube-root gauges make this block J_3 and both triangle Gram blocks $18I_3 - 3J_3$. Each pair can first be gauged to off-diagonal ± 3 . A negative pair has a symmetric eigenvector of eigenvalue 12; its two entries against every triangle column would need to sum to zero. Two cube roots cannot do so. All six pair blocks are thus positive.

Each 2×2 pair-to-pair block has the form

$$\begin{pmatrix} a & b \\ b & a \end{pmatrix}, \quad a, b \in \mu_3.$$

In symmetric and antisymmetric pair coordinates, write its entries as $S = a + b$ and $D = a - b$. The antisymmetric 6×6 block satisfies $DD^* = D^*D = 12I_6$. Every row and column has four nonzero entries of squared norm 3 and two zero entries. Correspondingly, an entry of S is $2a$ at a zero of D and is a negative cube root otherwise.

The triangle-to-pair columns and pair-to-triangle rows are balanced cube-root triples. Their Gram conditions force three of each of the two cube-root orbits. Pair gauges and permutations make the two groups identical within each orbit. The cross-Gram equations then say that every row and column of S sums to zero within each group of three. A group cannot contain two zeros of D , because $2a + 2b - c = 0$ has no cube-root solution. Since there are two zeros in total, each group contains exactly one. Also $2a - b - c = 0$ forces $a = b = c$.

It follows that each 3×3 block of S is $\alpha(3P - J_3)$ for a cube root α and a permutation matrix P . Orthogonality between the two row groups would require

$$\alpha(9R - 3J_3) + \beta(9T - 3J_3) = 0, \quad \alpha, \beta \in \mu_3, \quad R, T \in S_3.$$

All $3^2 \cdot 6^2 = 324$ possibilities are checked exactly and none vanishes. Equivalently, on the zero-sum plane a permutation would have to be a scalar equal to the negative of a cube root; the only scalar action of a permutation there is the identity. This completes the no-isolate boundary. The finite checks and the graph/spectrum reductions are in `programs/verify_q81.py`.

6 Independently recovered $Q = 99$ local certificates

The historical final types under examination are

$$A = 4K_2^- \sqcup K_3^+ \sqcup C_4^+, \quad B = 3K_2^- \sqcup K_3^+ \sqcup T_6,$$

where T_6 has a positive edge between its centers and two negative leaf edges at each center. Both specified full Grams have characteristic polynomial

$$(x - 21)^2(x - 18)^4(x - 15)^2(x - 12)^6(x - 9).$$

Their 15-eigenspaces are supported on the four cycle vertices or the four leaves, respectively. This makes the original residual shape conditions necessary: columns outside the relevant local block must be orthogonal to that localized eigenspace.

For each local pair the new verifier constructs the integer linear operator for $LZ - ZR = 0$, row-reduces it over $\mathbb{Q}$, assigns cube roots to the free matrix-entry coordinates, and reconstructs every pivot coordinate. It then checks membership in μ_3 and compares the entire resulting set with the original parameterization.

Pair	Variables	Rank	Free	Cube-root solutions	Compatible residuals
A/B	24	18	6	144	0
A/A	16	10	6	225	0
B/B	36	28	8	729	0

The raw free-coordinate enumerations have sizes 3^6 , 3^6 and 3^8 . This validates completeness of the *local* parameterizations and their contradictions. It does not validate an exhaustive reduction of all $Q = 99$ matrices to these two global types, nor the separate mixed-color size-13 cases. Those remain explicit obligations.

7 Record-level component bounds for $(14, 1)$

Assume the 14-row same-color internal support has no isolates. Let its connected components have sizes $v_i \geq 2$ and energy units $u_i \geq v_i - 1$, so $\sum v_i = 14$ and internal energy $e = 9 \sum u_i$. Write $c = Q - e \geq 42$ for the cross energy to the remaining row.

For a general component use determinant upper bound $U_{v_i}(9u_i)$ and largest-eigenvalue cap

$$L_i = 15 + \sqrt{\frac{18u_i(v_i - 1)}{v_i}}.$$

When $u_i = v_i - 1$, the component must be a norm-9 tree. All edge phases can be gauged away, and the determinant is exactly that of $15I + 3A_T$. Bipartiteness also gives the sharper cap $L_i = 15 + 3\sqrt{u_i}$. Rational upper brackets for every square root preserve the direction of the Schur bound.

The minimum squared norm on every cross entry is 3. Allocating all excess cross energy to the component with the largest cap only weakens the bound, and gives

$$\det G \leq \prod_i d_i \left(15 - \sum_i \frac{3v_i}{L_i} - \frac{c - 42}{\max_i L_i} \right), \quad (3)$$

with zero as an upper bound when the displayed Schur allowance is nonpositive. Here d_i is the general determinant envelope or the exact maximum over trees of that order.

The tree determinant is computed by its matching polynomial,

$$\det(15I + 3A_T) = \sum_j (-9)^j m_j(T) 15^{|T| - 2j},$$

using exact rooted-tree dynamic programming. All nonisomorphic trees of orders 2 through 14 are enumerated. At order 14, there are 3159 trees; the unique determinant maximizer is P_{14} , with value 16802420983158456. The same enumeration records the unique path maximizer at each smaller order.

All integer component abstractions are enumerated, including unrealizable ones; this enlarges the feasible set safely. Bound (3) is at most B for every internal energy allocation at each of the eight eligible energies 105, 114, $\dots$, 168. There are 36 allocations in total. At $Q = 105$ the no-isolate support is $7K_2$, and (3) yields

$$216^7 \left(15 - \frac{42}{18} \right) = 216^7 \frac{38}{3} = B.$$

At the other seven energies the computed maxima are strictly below B . This result does not rely on an unverified general “isolate exclusion” lemma; its no-isolate hypothesis remains explicit.

7.1 Why the old $Q = 150$ test was insufficient

The inspected historical script compares its branches to the old $Q = 153$ trace envelope. Its six size-14 expressions use

$$(e, c, L) = (63, 87, 25), (72, 78, 127/5), (81, 69, 261/10), \\ (90, 60, 27), (99, 51, 138/5), (108, 42, 563/20)$$

and evaluate $U_{14}(e)(15 - c/L)$. Five of the six expressions are greater than B , despite all being below the old envelope. Their ratios to B are approximately

$$1.00484, \quad 1.00383, \quad 1.00457, \quad 1.00587, \quad 1.00307, \quad 0.999981.$$

This is a certificate-strength gap, not a counterexample. The new component proof repairs this particular no-isolate branch. It still does not supply the missing size-13 state-pairing and isolate arguments, so the whole $Q = 150$ shell is not declared closed.

8 Reproducibility and remaining obligations

The entry point `run_all.py` executes every available verifier in a fresh subprocess and saves stdout, stderr, exit codes and timings. `--audit-only` requests completion of the audit. Without that option, unresolved global obligations cause exit code 2 even when every available local test passes. Source hashes are checked before execution; Python optimization mode, which disables assertions, is rejected.

The pinned numerical dependencies are SymPy 1.13.1 and NetworkX 3.5. Graph-atlas completeness and nonisomorphic-tree generation are trusted library inputs; this delivery is not a formal proof assistant development and does not independently formalize those graph generators. The external code-classification theorem is identified rather than silently assumed to have been recomputed.

The main missing obligations are the record-level size-13 defect and local inverse arguments; completeness of the large support/phase catalogues; the cubic-majorant, high-triangle and Fourier branches; the $Q = 144, 150$ state-pairing certificates; and the upstream sparse/spectral coverage of the historical tail closures. The detailed per-shell list is in `report/remaining_obligations.md`. The aggregate code never replaces a missing proof by a boolean or a fixed expected answer.

All reported fractions, characteristic polynomials, enumeration counts and partial bounds are saved in `results/`. The original archive's ledger remains preserved under `original/`, but its labels do not control the new status table. A claim of global maximality requires every unresolved row below to acquire a complete record-level proof.

A Complete eligible-shell coverage table

Closed means a complete reconstructed shell argument passed. **Closed, cited input** additionally uses $B_3(15, 10) = 12$. **Unverified** means that a complete global argument for that shell has not been recovered and checked. It does not assert falsity or exhibit a counterexample.

Q	Audit status	Evidence / remaining obligation
0	Closed	Exact no-leaf catalogue, norm sieve and two-sided kernel projection.
6	Closed	Color reduction and minimum cross energy 42.
9	Closed	Exact no-leaf catalogue, norm sieve and two-sided kernel projection.
15	Closed	Color reduction and minimum cross energy 42.

Continued on next page

Q	Audit status	Evidence / remaining obligation
18	Closed	Exact no-leaf catalogue, norm sieve and two-sided kernel projection.
24	Closed	Color reduction and minimum cross energy 42.
27	Closed	Exact no-leaf catalogue, norm sieve and two-sided kernel projection.
33	Closed	Color reduction and minimum cross energy 42.
36	Closed	Exact no-leaf catalogue, norm sieve and two-sided kernel projection.
42	Closed (cited in-put)	Color/energy bound plus $B3(15,10)=12$.
45	Closed	Exact no-leaf catalogue, norm sieve and two-sided kernel projection.
51	Closed (cited in-put)	Color/energy bound plus $B3(15,10)=12$.
54	Closed	Exact no-leaf catalogue, norm sieve and two-sided kernel projection.
60	Closed (cited in-put)	Full small mixed support/cross enumeration and eigen-projection PSD; $B3(15,10)=12$.
63	Closed	Exact no-leaf catalogue, norm sieve and two-sided kernel projection.
69	Closed (cited in-put)	Full small mixed support/cross enumeration and eigen-projection PSD; $B3(15,10)=12$.
72	Closed	Exact no-leaf catalogue, norm sieve and two-sided kernel projection.
78	Closed (cited in-put)	Full small mixed support/cross enumeration and eigen-projection PSD; $B3(15,10)=12$.
81	Closed (cited in-put)	Isolate support catalogue and exact $K3+6K2$ block obstruction; $B3(15,10)=12$.
87	Unverified	Missing: Record-level size13 defect/local inverse reductions and other remaining support branches; a lower historical comparison target is insufficient.
90	Unverified	Missing: Record-level size13 defect/local inverse reductions and other remaining support branches; a lower historical comparison target is insufficient.
96	Unverified	Missing: Record-level size13 defect/local inverse reductions and other remaining support branches; a lower historical comparison target is insufficient.
99	Unverified	Partial: All AB/AA/BB local μ_3 intertwiners independently recovered; 144/225/729 solutions, zero compatible residuals. Missing: Exhaustive reduction to A/B and mixed-color branches, especially size13 internal energy9/18.
105	Unverified	Partial: (14,1) no-internal-isolate branch completely bounded by B over all component allocations.; An explicit matrix attains B; the 7K2 friendship support family has exact upper bound B. Missing: Complete record-level proof for every color, support, and phase case.
108	Unverified	Missing: Record-level size13 defect/local inverse reductions and other remaining support branches; a lower historical comparison target is insufficient.

Continued on next page

Q	Audit status	Evidence / remaining obligation
114	Unverified	Partial: (14,1) no-internal-isolate branch completely bounded by B over all component allocations. Missing: Record-level size13 defect/local inverse reductions and other remaining support branches; a lower historical comparison target is insufficient.
117	Unverified	Missing: Record-level size13 defect/local inverse reductions and other remaining support branches; a lower historical comparison target is insufficient.
123	Unverified	Partial: (14,1) no-internal-isolate branch completely bounded by B over all component allocations. Missing: Record-level size13 defect/local inverse reductions and other remaining support branches; a lower historical comparison target is insufficient.
126	Unverified	Missing: Complete polynomial-majorant/high-triangle orbits, Fourier branches and mixed-color record-level reductions.
132	Unverified	Partial: (14,1) no-internal-isolate branch completely bounded by B over all component allocations. Missing: Record-level size13 defect/local inverse reductions and other remaining support branches; a lower historical comparison target is insufficient.
135	Unverified	Missing: Complete polynomial-majorant/high-triangle orbits, Fourier branches and mixed-color record-level reductions.
141	Unverified	Partial: (14,1) no-internal-isolate branch completely bounded by B over all component allocations. Missing: Record-level size13 defect/local inverse reductions and other remaining support branches; a lower historical comparison target is insufficient.
144	Unverified	Missing: Complete K6 support catalogue and record-level state-pairing certificates.
150	Unverified	Partial: (14,1) no-internal-isolate branch completely bounded by B over all component allocations. Missing: Record-level (13,2) local Schur/state-pairing and (14,1) isolate exclusion; the no-isolate branch has now been reconstructed.
153	Unverified	Missing: Full upstream sparse/spectral/support reductions of the tail certificate, including all-same/mixed residuals. The historical closure claim has not been fully replayed here.
159	Unverified	Partial: (14,1) no-internal-isolate branch completely bounded by B over all component allocations. Missing: Full upstream sparse/spectral/support reductions of the tail certificate, including all-same/mixed residuals. The historical closure claim has not been fully replayed here.
162	Unverified	Missing: Full upstream sparse/spectral/support reductions of the tail certificate, including all-same/mixed residuals. The historical closure claim has not been fully replayed here.

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<i>Q</i>	Audit status	Evidence / remaining obligation
168	Unverified	Partial: (14,1) no-internal-isolate branch completely bounded by B over all component allocations. Missing: Full upstream sparse/spectral/support reductions of the tail certificate, including all-same/mixed residuals. The historical closure claim has not been fully replayed here.

B Execution record

Module	Fresh replay result
verify_arithmetic_tests	Arithmetic cross-checks PASS: 60 matrices; 94 trees
verify_foundations	Foundations PASS: det pair (604661760, 241864704) B 277868041444786176 Q 105 tail/B 0.9944244061347672
verify_low_energy	Q 0 VERIFIED_CLOSED assemblies 1 norm survivors 0 final 0; Q 9 VERIFIED_CLOSED assemblies 0 norm survivors 0 final 0; Q 18 VERIFIED_CLOSED assemblies 0 norm survivors 0 final 0; Q 27 VERIFIED_CLOSED assemblies 0 norm survivors 0 final 0; Q 36 VERIFIED_CLOSED assemblies 1 norm survivors 0 final 0; Q 45 VERIFIED_CLOSED assemblies 1 norm survivors 1 final 0; Q 54 VERIFIED_CLOSED assemblies 7 norm survivors 1 final 0; Q 63 VERIFIED_CLOSED assemblies 15 norm survivors 1 final 0; Q 72 VERIFIED_CLOSED assemblies 58 norm...
verify_mixed_small	mixed Q 60 VERIFIED_WITH_CITED_CODE_BOUND states 36 norm 3 residual 0; mixed Q 69 VERIFIED_WITH_CITED_CODE_BOUND states 1008 norm 26 residual 0; mixed Q 78 VERIFIED_WITH_CITED_CODE_BOUND states 17424 norm 201 residual 0
verify_q81	Q81 VERIFIED_WITH_CITED_CODE_BOUND isolate states 22 -> 0 non-isolate spectra 4 -> 0
verify_q99_independent	Q99 local rederivation PASS: 144,225,729 solutions; all residual survivors zero.
verify_component_bounds	Q 105 (14,1) no-isolate: 1 / 1 allocations; max/B 1.0; Q 114 (14,1) no-isolate: 2 / 2 allocations; max/B 0.9620720360256458; Q 123 (14,1) no-isolate: 3 / 3 allocations; max/B 0.9566417785303339; Q 132 (14,1) no-isolate: 4 / 4 allocations; max/B 0.9579734857152806; Q 141 (14,1) no-isolate: 5 / 5 allocations; max/B 0.9636743412482026; Q 150 (14,1) no-isolate: 6 / 6 allocations; max/B 0.9727713273053784; Q 159 (14,1) no-isolate: 7 / 7 allocations; max/B 0.9846300571620813; Q 168 (14,1) no-isolate: 8 / 8 allocations...
audit_all_shells	Global coverage: 19 closed; 19 unverified. Global maximality: False

All listed modules returned exit code 0 in the reconstructed replay. The aggregate shell audit reports 19 closed shells, 19 unverified shells, and `global_maximality_verified = false`. This is intentionally treated as an incomplete global proof.

The immutable-source manifest and the full delivered snapshot manifest have different purposes. A replay updates log timings and result files; the source manifest remains the

pre-execution check. The original upload hash is in `source_provenance.json`. No complete repository checkout or missing remote proof file is implicitly included in these manifests.

References

- [1] T. Todorov and G. Bogdanova. *Ternary equidistant codes of length 11 through 15*. Journal of Mathematical and Computational Science 10 (2020), 2713–2721. Proposition 14 and Table 1. <https://doi.org/10.28919/jmcs/4964>.
- [2] Order-15 research materials in *order55-binary-circulant-maxdet*, owner dongxuelian2. Examined main ref 827672a2c60dad7064c7955d9c460261d866d25e and research ref 007af7d8f7f644396aee27dd87c68c8f4169efd5. Paths and recovery limits are documented in `recovered/source_notes.md`.
- [3] User-supplied `order15_mu3_Q87-150_bundle.zip`. Sixteen recovered files, including the final ledger, proof digest, process notes and standalone $Q = 99$ verifier. Preserved without adopting unsupported completion labels.